

Optimisation of deficit irrigation strategies under climate change conditions based on artificial intelligence

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Highlights:

- Assesses future irrigation requirements under climate change scenarios.
- Integrates climate projections to evaluate deficit irrigation performance.
- Uses advanced optimisation algorithms to maximise yields.
- Provides a multi-criteria framework for selecting optimal crop patterns.

Abstract: This paper presents a novel modelling framework that integrates artificial intelligence (AI) tools with deficit irrigation (DI) management. Different levels of DI were applied to summer and winter crops, with six scenarios assessed for each season. Four metaheuristic algorithms – LSHADE (Success-History based Adaptive Differential Evolution with Linear Population), PSO (Particle Swarm Optimisation), GWO (Grey Wolf Optimiser), and NRO (Nuclear Reaction Optimisation) – were implemented and automated using Python-based models to identify optimal solutions. The optimisation model aims to maximise the project's total net profit by determining the most efficient allocation of land and water resources under DI conditions. For the summer season, the findings showed that the best crop pattern was obtained in scenario S2, the balanced traditional scenario, using the LSHADE algorithm. This scenario increased financial profit by 29%, used 89% of the land, and applied 12.5% deficit irrigation. The winning crop pattern consisted of 25% maize, 5% tomato, 5% cucumber, 4.23% aubergine, 10% watermelon, and 40% maize forage. For the winter season, the optimal pattern was obtained in scenario S6 (water-stressed) using the LSHADE algorithm. Financial profit increased by 44%, land use reached 100%, and DI was 12.5%. The most suitable agricultural pattern included 45% wheat, 30% barley, 5% potatoes, 3% spinach, and 7% alfalfa.

The findings confirm that AI-driven optimisation effectively supports DI management. Advanced metaheuristic algorithms should be integrated into irrigation planning, and their adoption should be supported through simple decision-support tools for farmers.

Keywords: Abu Ghraib Irrigation Project, crop pattern, deficit irrigation (DI) management, optimisation algorithm, smart irrigation

INTRODUCTION

Climate change has become a serious global issue, evident in rising temperatures, more frequent droughts, and changes in rainfall frequency. Iraq and many other regions, especially arid and semi-arid areas, are particularly affected, with global warming adversely affecting water security and agriculture. Global warming has reduced available water resources, widening the gap between supply and demand. Since agriculture accounts for more than 70% of water consumption in many countries, it is particularly vulnerable to these challenges, requiring a reevaluation of water management policies. Recent studies confirm that temperatures in Iraq are increasing at rates higher than the global average, and climate models indicate that this trend will continue in the coming years (Saleh, Oleiwi and Lateef, 2023; Rasheed, Al-Khafaji and Alwan, 2024). Such warming intensifies evapotranspiration, thereby increasing crop water demand (CWD) and reducing crop water availability (CWA), with significant implications for water stress and food security (Adamo *et al.*, 2022). Rising temperatures associated with climate change reduce renewable water resources and alter rainfall patterns. This makes it difficult to cultivate entire agricultural areas, especially in arid regions that rely mainly on irrigation (Hasan and Abed, 2023; Talib and Laffta, 2024). Deficit irrigation (DI) has emerged as a critical management strategy in this context. The relationship between DI and yield reduction (YR) varies according to crop species and sensitivity at different growth stages, making it essential to balance productivity with water conservation under water-limited conditions (English, 1990). Optimisation techniques are effective in determining optimal irrigation levels that maintain production under water-scarce conditions. DI also supports the management of crop patterns that account for drought resistance, water-use efficiency, and economic returns. This approach is supported by multi-objective optimisation models that identify optimal crop and water allocations to maximise project benefits and improve food security. Given these challenges, artificial intelligence (AI) techniques have proven to be effective tools for smart and efficient water resource management. AI can learn from data and make better decisions that adapt to changing conditions. According to Abdulameer *et al.* (2025), AI has proven effective in predicting water demand by determining optimal irrigation times and amounts and predicting how plants will respond to water stress. Studies indicate that the use of AI in agriculture may help reduce water consumption by up to 30% in certain situations (Mohammed, Hamdoun and Sagheer, 2023; Wei *et al.*, 2024). The production and cost functions should be specified for the intended crop to determine the optimal use of water under land and water constraints (Hammouch *et al.*, 2024).

Many studies have benefited from the use of AI in water resource management, employing machine-learning algorithms such as linear regression and neural networks to estimate soil moisture and forecast irrigation requirements (Goap *et al.*, 2018; Elavarasan and Vincent, 2020). Other research has employed reinforcement-learning algorithms to develop intelligent models that learn effective irrigation policies over time (Saikai, Peake and Chenu, 2023; Aribisala, Ghorri and Cavalcante, 2025). However, most of these studies focus on optimising irrigation timing or reducing water consumption without addressing economic

aspects such as prices and productivity or considering DI as an adjustable water-deficit ratio. Therefore, applications of advanced metaheuristic optimisation algorithms such as PSO, NRO, LSHADE, and GWO remain limited, particularly in models that simulate real-world farm management or actual irrigation projects. Such models require the integration of productivity, prices, DI ratios, and water consumption within a unified objective function aimed at maximising economic returns. The current paper aims to develop an optimisation model that determines the optimal distribution of agricultural areas and the optimal DI level for each crop based on available resources, including the total quantity of water and crop market prices. This model was applied to the Abu Ghraib Irrigation Project (AGIP), one of the largest agricultural projects west of Baghdad. AGIP is a strategic agricultural area facing significant challenges in water resource management and increasing water stress under the influence of climate change.

The main objectives of this study are: (1) to assess the impacts of climate change on crop water requirements in the study area, (2) to evaluate the application of DI strategies, (3) to develop an AI-based optimisation function that achieves an optimal balance between economic return and water-use efficiency, and (4) to compare the performance of advanced metaheuristic algorithms to identify the most efficient and robust solution. The research problem addressed in this paper is how to determine an optimal distribution of crops within the agricultural plan of AGIP in Iraq and establish DI levels for each crop that achieve the best possible productivity, or at least minimise productivity losses, under DI conditions. This requires an objective function that accounts for the available area for each crop, the impact of DI on production, crop price, and water consumption. Four algorithms were employed for this purpose to search for the best combination of crops, areas, and DI ratios that maximises economic returns and minimises the negative impact of water scarcity. The novelty of this study lies in integrating a multi-objective, multi-criteria optimisation framework with advanced AI-based metaheuristic algorithms under future climate-change scenarios. This addresses a clear research gap in the existing literature, where most previous studies focused on single-objective optimisation or short-term water-productivity analysis. This research simultaneously evaluates economic return, water saving, crop sensitivity, and algorithmic performance within a unified decision-support framework.

MATERIALS AND METHODS

STUDY AREA DESCRIPTION

Climatic data collection

The climate of Iraq is generally considered arid to semi-arid, with long, hot, dry summers in which temperatures reach 50°C and short, cold winters with low rainfall. Annual rainfall ranges from 70 mm to 200 mm (Iraqi Meteorological Organization and Seismology, 2024). Frequent northwesterly winds further intensify these conditions. Climate data collected by the Baghdad Meteorological Station for the period 1980–2023 were used in this paper to generate different climate scenarios using the Long Ashton Research Station Weather Generator (LARS-WG). The dataset includes daily temperature (°C), relative humidity (%),

wind speed ($\text{m}\cdot\text{s}^{-1}$), sunshine duration ($\text{h}\cdot\text{day}^{-1}$), and precipitation (mm). Temperature and radiation data were verified using NASA's POWER database (<https://power.larc.nasa.gov>), while precipitation data were verified using the CLIMATE ENGINE platform and CHIRPS satellite data at a high spatial resolution of $0.05^\circ \times 0.05^\circ$. The historical period 1993–2023 was selected because it represents a standard duration for assessing climate variability and helps reduce the influence of short-term fluctuations. These websites are preferred for research because of their extensive geographical coverage and accurate temporal coverage. Saleh, Olewi and Lateef (2023) and Ahmed and Hadid (2025) emphasise their importance in providing free, up-to-date information supported by advanced technologies such as satellites.

Location and irrigation network

Iraq is situated within the arid and semi-arid continental zones of the Middle East, between latitudes $29.5\text{--}37.22^\circ\text{N}$ and longitudes $38.45\text{--}48.45^\circ\text{E}$. The AGIP is located 35 km southwest of Baghdad and is bordered by the Saqlawiyah and Radhwaniyah Irrigation Projects. It covers an area of 515 km^2 and is irrigated by a combined canal system. The main canal is an 18 km-long, concrete-lined channel with a discharge of $28\text{ m}^3\cdot\text{s}^{-1}$. The climate of the AGIP is characterised by cold, low-rainfall winters and hot, dry summers, making the area important for study in light of changing climatic conditions (Ministry of Water Resources, 2014; Abdullah, Laue and Al-Ansari, 2019). The AGIP is one of the largest and most important irrigation projects in the central region of Iraq (Fig. 1). Its strategic location within an arid region increases the importance

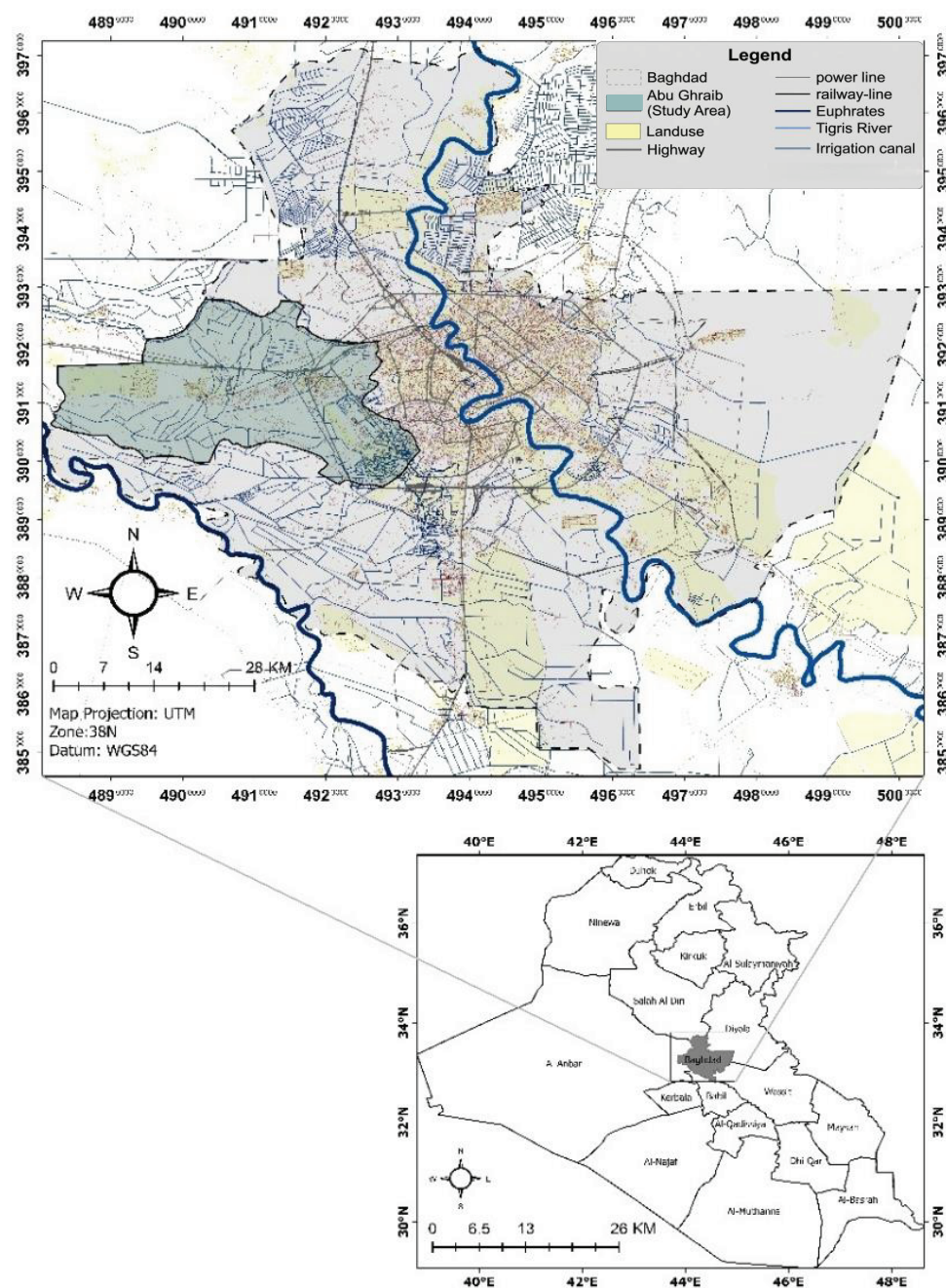


Fig. 1. Regional map showing the relative location of the Abu Ghraib Irrigation Project (AGIP) within the Governorate of Baghdad and Iraq; source: own elaboration

of its study and its significance for national food security. The project depends directly on the Euphrates River, whose water supplies have decreased in recent years.

ESTIMATION OF CROP WATER REQUIREMENTS

The CROPWAT 8.0 (FAO) computer program was used to estimate crop water requirements and calculate water productivity and crop yields under *DI* strategies supporting the objective function. It calculates reference evapotranspiration (ET_o) from integrated soil, crop, and weather data (Allen *et al.*, 1998; Al-Haddad and Bakr, 2013), providing reliable input variables for AI models aimed at balancing water savings and yield. Climate variables, including maximum and minimum temperature ($^{\circ}\text{C}$), wind speed ($\text{km}\cdot\text{day}^{-1}$), sunshine duration, altitude, relative humidity (%), latitude, longitude, and rainfall, were collected, averaged, and entered into the model. Crop evapotranspiration (ET_c) is computed by multiplying ET_o by the crop coefficient (K_c) using the Penman–Monteith equation (Eq. 1).

$$ET_c = K_c \cdot ET_o \quad (1)$$

CLIMATIC SIMULATION

The LARS-WG model is a stochastic weather generator widely used in climate research and hydrological modelling. It was used to simulate synthetic climate data for the AGIP site under historical conditions for the period 1993–2023 and to generate a long time series of daily and monthly weather parameters for 2026. The model generates and simulates a range of weather parameters, including minimum and maximum temperatures, wind speed, and precipitation. These parameters are important inputs for different hydrological and environmental models (Abidalla and Abed, 2025). The RCP 4.5 scenario was selected as a moderate stabilisation pathway aligned with global mitigation strategies (Hassan *et al.*, 2023). HadGEM3 GC31-LL provides reliable climate projections due to its comprehensive representation of climate variables and participation in the Coupled Model Intercomparison Project (CMIP6) (Michaelides, 2023). The Shared Socioeconomic Pathway 245 (SSP 245) represents a middle socioeconomic scenario, making it suitable for studying future climate impacts (Zhai, Zhuo and Wang, 2024).

DEFICIT IRRIGATION LEVELS AND YIELD REDUCTION RESPONSES

The yield reduction (*YR*) values were manually derived using CROPWAT 8.0 software based on different *DI* levels ranging from 0%, 5%, and 10% up to 50%. For example, a *DI* level of 5% was applied, and the percentage reduction in water was calibrated for each crop growth stage across all phenological stages. The upper deficit limit of 50% was selected because yield losses usually become very high beyond this level for most field and vegetable crops, according to previous studies. Including this level allows system performance to be evaluated under severe water stress while keeping the scenarios realistic for agricultural conditions. Model-predicted yields under each *DI* level were then used to calculate the corresponding *YR* values. This procedure was implemented for each crop at multiple *DI* levels to assess crop sensitivity to water deficit. The relationships between *YR* and *DI*

were obtained and expressed using quadratic equations. These equations were then used in the objective function as decision variables to calibrate the impact of *DI* on yields and, consequently, on final profit.

OPTIMISATION TECHNIQUE BASED ON ARTIFICIAL INTELLIGENCE

Estimating irrigation requirements and crop yields is essential for farmers and policymakers. Farmers depend on accurate predictions of irrigation needs to support agricultural crop yields. This approach contributes to reducing water consumption and increasing crop yields. However, forecasting crop yields precisely is also difficult. Accurate estimates of irrigation requirements and crop yields are also beneficial to governments because they support the development of plans for irrigation water needs. In this paper, four AI-based metaheuristic optimisation algorithms were used to solve the irrigation and crop-pattern optimisation problem: Particle Swarm Optimisation (PSO), Nuclear Reaction Optimisation (NRO), Success-History based Adaptive Differential Evolution with Linear Population (LSHADE), and Grey Wolf Optimiser (GWO).

These algorithms were chosen for their proven efficiency and adaptability in optimising agricultural and water-resource management. PSO is a population-based optimisation algorithm that searches a large region of the solution space simultaneously. It is inspired by the social behaviour of birds and fish schooling (Kennedy and Eberhart, 1995). NRO is a population-based algorithm designed for extensive exploration and avoidance of local optima. It is effective for complex and multimodal problems in which solution landscapes are highly irregular (Sivasundari, Rao and Raju, 2019). LSHADE is an advanced version of differential evolution that adapts its parameters during the search process. It performs well in global optimisation tasks and is suitable for large-scale agricultural datasets (Nebro, Durillo and Coello Coello, 2013). GWO mimics the hunting behaviour of grey wolves. It is widely used to improve crop yield and optimise crop planting density, for example by optimising sprinkler placement in a potato field to maximise yield.

OPTIMISATION MODEL STRUCTURE

This model aims to maximise net benefit. Its decision variables include the area allocated to each crop and the applied *DI* level. The model also includes several constraints, such as the total available area, the total amount of water entering the project, and the minimum and maximum allowable areas for each crop. The model used four optimisation algorithms: PSO, NRO, LSHADE, and GWO. Each algorithm was tested separately for ten iterations using Python to ensure the reliability of the results. In each experiment, 500 automatic iterations were performed for each algorithm. Six different scenarios were applied to the summer season and six to the winter season, taking into account differences in *DI* levels and area allocations. This model provides a comprehensive framework for various fields of irrigation management.

Decision variables

Crop area (*A*). The area occupied by crop *A* is considered a primary decision variable in the optimisation function. It represents the area allocated to cultivation of each crop.

Determining the optimal areas enables effective use of available land and increases economic returns, especially under constraints on total land availability. Total yield and net benefit are directly influenced by the distribution of area among different crops.

Yield reduction (YR) decision variable. The *DI* strategy was applied at several levels to determine the YR ratio for each crop separately, reflecting crop sensitivity to water stress. This sensitivity varies by crop type; therefore, *DI* was applied at seven levels (0%, 5%, 10%, 15%, 20%, 25%, and 50%) to observe its impact on productivity throughout the crop's lifespan. YR data were generated using CROPWAT, with manual calculations of water requirements at each growth stage. Data obtained from CROPWAT were then statistically fitted using Table Curve to represent the relationships accurately. The resulting quadratic functions for each crop were subsequently used as decision variables in the objective function of the optimisation model.

Constant parameters used in the optimisation model for both winter and summer crops are presented in Table 1. These parameters are the actual cultivated area per hectare, market price in USD per kilogram, and average yield in kilograms per hectare. The values were obtained from the Ministry of Water Resources report (MoWR, 2014) and serve as constant input data in the function.

Objective function of the optimisation model

Mathematical formulation:

$$Z = \sum_{i=1}^n [A_i \cdot Y_{\max} (1 - YR_i(DI_i)) \cdot P_i] \quad (2)$$

where: Z = the total net benefit (USD), A_i = area allocated to crop i (ha), $Y_{\max}$ = maximum yield (productivity) of crop i ($\text{kg}\cdot\text{ha}^{-1}$), $YR_i(DI_i)$ = yield reduction factor of crop i as a function of deficit irrigation levels, DI_i = deficit irrigation level for crop i (% of full irrigation), P_i = price of crop ($\text{USD}\cdot\text{kg}^{-1}$).

Constraint factors

Water allocation constraint for each crop. The annual water requirement (Wu_i) allocated to each crop was expressed as a linear function of the *DI* levels (0%, 5%, 10%, 15%, 20%, 25%, and 50%).

$$Wu_i = a_i \cdot DI_i + b_i \quad (3)$$

where: a_i , b_i = coefficients derived from CROPWAT-based evapotranspiration (*ET*) calculations by analysing variation in crop water consumption across different *DI* levels, which showed a progressive decrease in water requirements under water stress.

Total water availability constraint. The total water used across all crops and areas must not exceed the available water resources:

$$\sum_{i=1}^n A_i \cdot Wu_i \leq W_{\text{total}} \quad (4)$$

where: W_{total} = total irrigation water available.

Total annual water diverted to the project across all optimisation scenarios was set to $20 \text{ m}^3\cdot\text{s}^{-1}$.

Land area constraint of each crop. Minimum and maximum bounds limit the cultivated area for each crop:

$$A_{\min} \leq A_i \leq A_{\max} \quad (5)$$

where: $A_{\min}$ = minimum required area for crop i (to ensure food security), $A_{\max}$ = maximum allowable area for crop i .

After defining the land-area constraints for each crop, the minimum and maximum area limits were set for all six scenarios considered in this paper. The lower and upper allowable bounds of the cultivated area are presented in Table 2. Six scenarios were selected to represent the most plausible crop-allocation strategies in AGIP. S1 reflects the actual current scenario, while S2 represents the balanced traditional pattern and S3 prioritises fodder crops to meet rising livestock-feed demand. Scenarios S4 and S5, representing moderate diversification and vegetable-intensive production, simulate shifts towards higher-value crops. Finally, S6 represents water-stressed conditions to capture the expected impacts of reduced irrigation supply under climate change. These estimated area limits are based on practical considerations, crop importance, and strategic factors.

Total land area constraint. The sum of all cultivated areas must not exceed the total available agricultural land (A_{total}) – Equation (6).

$$A_{\text{total}} = \sum_{i=1}^n A_i \quad (6)$$

where: n = number of crops.

RESULTS AND DISCUSSION

CROP SENSITIVITY AND RELATIONSHIPS OF YIELD REDUCTION AND DEFICIT IRRIGATION

The relationships between yield reduction (YR) and deficit irrigation (*DI*) are shown in Table 3 as quadratic equations generated for the studied winter and summer crops. The high

Table 1. Constant parameters for winter and summer crops in Abu Ghraib Irrigation Project

Winter crop	Actual area (ha)	Price ($\text{USD}\cdot\text{kg}^{-1}$)	Yield ($\text{kg}\cdot\text{ha}^{-1}$)	Summer crop	Actual area (ha)	Price ($\text{USD}\cdot\text{kg}^{-1}$)	Yield ($\text{kg}\cdot\text{ha}^{-1}$)
Wheat	7,755.625	0.53	1,761.2	maize	1,810.625	0.34	2434.0
Barley	712.500	0.44	1,102.4	tomato	482.500	0.31	16000.0
Potato	2,006.250	0.42	12,719.2	cucumber	651.250	0.45	8345.6
Spinach	146.875	0.42	9,977.2	aubergine	250.000	0.41	16720.0
Alfalfa	109.375	0.34	28,908.0	watermelon	260.625	0.35	17400.0
Forage mixture	214.375	0.44	16,445.2	maize forage	189.375	0.34	2434.0

Source: own elaboration.

Table 2. Minimum and maximum allowable area ratios (%) allocations for each crop

Crop	Area ratio at scenario											
	S1 actual scenario		S2 balanced traditional		S3 fodder priority		S4 moderate diversification		S5 vegetable-intensive		S6 water-stressed	
	min.	max.	min.	max.	min.	max.	min.	max.	min.	max.	min.	max.
Winter crop												
Wheat	30	45	20	40	10	25	15	35	10	20	25	45
Barley	15	25	15	30	10	20	10	25	5	10	15	30
Potato	5	10	5	15	5	10	10	20	20	30	2	5
Spinach	1	5	3	10	2	5	5	10	15	25	1	3
Alfalfa	10	15	10	20	20	30	10	20	10	15	5	10
Summer crop												
Forage mixture	10	15	10	20	20	30	10	15	5	10	5	10
Maize	20	35	15	25	15	30	10	20	20	35	25	40
Tomato	10	20	5	10	10	20	20	30	5	10	10	15
Cucumber	5	15	3	8	5	10	10	20	3	5	5	10
Aubergine	5	15	3	8	5	10	10	20	3	5	5	10
Watermelon	10	20	5	10	10	15	10	15	5	10	10	15
Maize forage	10	20	30	40	10	15	5	10	10	15	15	20

Source: own elaboration.

Table 3. Yield reduction (YR), sensitivity, and characteristics of crops under deficit irrigation (DI)

Crop	$YR = f(DI)$	R^2	Sensitivity to deficit irrigation	Crop characteristic influencing DI impact
Winter crop				
Wheat	$YR = 0.0012DI^2 + 0.0107DI - 0.0217$	0.99	maintains yield under high DI	deep root system and flexible growth stages
Barley	$YR = 0.0022DI^2 - 0.0517DI + 0.1484$	0.98	moderate yield loss at mid DI levels	waxy leaves and a short growing season
Potato	$YR = 0.0017DI^2 + 0.066DI - 0.2833$	0.98	high yield reduction under DI	shallow roots and high water demand during tuber formation
Spinach	$YR = 0.0005DI^2 + 0.1199DI - 0.3919$	0.97	moderate yield loss at mid DI levels	soft leaves, high evapotranspiration rate, shallow root depth
Alfalfa	$YR = 0.0012DI^2 + 0.0605DI - 0.1603$	0.99	maintains yield under high DI	continuous biomass production, high seasonal water needs
Summer crops				
Forage mixture	$YR = 0.0017DI^2 + 0.0064DI + 0.0241$	0.99	maintains yield under high DI	stomatal closure ability, no dependence on fruiting
Maize	$YR = 0.0006DI^2 + 0.0532DI - 0.1121$	0.99	maintains yield under high DI	sensitive during flowering and grain filling, large leaf area increases evapotranspiration
Tomato	$YR = 0.0031DI^2 - 0.029DI + 0.0677$	0.99	high yield reduction under DI	shallow root system, sensitivity during fruit setting
Cucumber	$YR = -0.0017DI^2 + 0.2608DI - 1.6935$	0.96	steep YR-DI curve, $R^2 = 0.96$	high water content in fruit, succulent tissue, and shallow roots

cont. Tab. 3

Crop	$YR = f(DI)$	R^2	Sensitivity to deficit irrigation	Crop characteristic influencing DI impact
Aubergine	$YR = 0.0004DI^2 + 0.0797DI - 0.3021$	0.91	moderate yield loss at mid DI levels	high transpiration due to a large canopy, extended fruiting period
Watermelon	$YR = 0.001DI^2 + 0.0352DI - 0.1314$	0.99	moderate yield loss at mid DI levels	large fruit size, high water content, shallow root system, large leaves
Maize forage	$YR = 0.0002DI^2 + 0.0056DI - 0.0438$	0.93	maintains yield under high DI	short growing cycle, waxy leaves, early harvest

Source: own study.

R^2 values (0.91–0.99) confirmed the accuracy of these relationships. Crop sensitivity ranged from low (e.g., wheat, alfalfa, and maize forage) to high (e.g., potato, tomato, and cucumber), depending on traits such as root depth, growth cycle, and water demand. These findings are consistent with previous studies (Fereses and Soriano, 2007; Geerts and Raes, 2009; Saeed *et al.*, 2021), which reported that crop sensitivity to DI varies according to physiological and morphological characteristics. These equations were developed for incorporation into the objective function of the optimisation model. Any change in the DI level directly affects the resulting YR and, consequently, the overall net profit calculated by the model.

In winter, the YR values of potato (15%), spinach (14.4%), and alfalfa (13%) were the highest among the studied crops. This may be attributed to their physiological characteristics and sensitive growth stages, which are strongly affected by water stress, limited water uptake from shallow soil layers, and the high water content of their edible parts, increasing yield losses under deficit irrigation. In contrast, the average YR values of mixed forage (9.6%), wheat (10.2%), and barley (7%) indicated greater tolerance to DI , with higher yields maintained due to low transpiration rates and deep roots. Their higher tolerance may also be explained by genetic adaptation to semi-arid environments, lower dependence on intensive irrigation, and the ability to partially recover from short periods of water stress due to relatively flexible growth cycles (Blum, 2010). For the summer plan, the YR values of cucumber (16.4%), tomato (12.6%), and watermelon (11%) were the highest among the studied crops because of their high transpiration rates, high water content, and shallow roots. Their higher sensitivity may also be related to vulnerability during flowering and fruit-setting stages, the combined effect of high temperature and water stress in summer, and the strong dependence of marketable yield on fruit size and quality. In contrast, the YR values of aubergine (10%), maize (9%), and maize forage (1.6%) reflect high tolerance to DI . In addition to leaf structure and root depth, this higher tolerance may also be associated with relatively high water-use efficiency, partial recovery capacity after short-term stress, and a yield structure that is less sensitive to water-related quality reductions.

COMPARATIVE ANALYSIS OF OPTIMISATION ALGORITHMS

The convergence plots created and analysed in Python show how GWO, LSHADE, NRO, and PSO performed across the summer-plan scenarios over 500 iterations (Fig. 2, scenarios S1–S6). LSHADE was the fastest and most successful algorithm in

achieving higher total benefits, followed by GWO, demonstrating their efficiency in solving irrigation-optimisation problems. NRO and PSO were slower or less stable and sometimes generated negative or unrealistic results. The findings identify LSHADE as the best choice for maximising total benefit under different conditions. Figure 3 shows the performance of the algorithms in the winter-plan scenarios, following the same approach.

Four algorithms are compared in terms of total utility and DI rate in Table 4. LSHADE achieved the highest utility, with complete stability and no variation in the results, while the DI rate remained constant at 12.5%. GWO showed behaviour close to that of LSHADE but with minimal variation and greater flexibility in DI rates. NRO and PSO showed greater variation in the results.

OPTIMISATION ANALYSIS OF CROP PATTERNS AND PROFITABILITY

The results reveal clear variation in total benefits, land-use distribution, and DI rates across the different summer scenarios (Tab. 5). Scenario S2 using LSHADE achieved the highest total benefit of USD 13,800.12, representing a 29.09% increase in profit, with 89.23% land use and a moderate average DI rate of 12.5%. In this scenario, allocating 25% of the area to maize and 40% to green forage reflects a suitable strategy for balancing food production with animal-feed demand and supporting food security. This was considered the winning scenario. The crop pattern in S2 also included moderate proportions of important crops, including tomato (5%), cucumber (5%), aubergine (4.23%), and watermelon (10%), diversifying income and enhancing the resilience of the production system against yield fluctuations. The results of this paper are consistent with those of Bulukazari *et al.* (2022), who showed that improving crop distribution using advanced irrigation models may increase profitability and water-use efficiency even under DI conditions in the Caspian Plain. A reduction in Wu of up to 20% did not result in large profit losses but instead increased economic productivity by up to 18% in some scenarios. For the winter scenarios, the average DI rate was consistently 12.5%, with 100% land use in all cases. Scenario S6 using LSHADE achieved the highest total benefit, reaching USD 6,955.65, with a 43.96% increase in total profit. It allocated higher proportions to wheat (45%) and barley (30%) – Table 5. This strategy focused on staple grain crops to maximise productivity. Scenario S6 is closely aligned with the results of Khoshsirat *et al.* (2022), where optimal DI levels maintained wheat and barley yields and significantly improved water-use efficiency. Similarly, Wang *et al.* (2023) demonstrated that carefully calibrated irrigation and nitrogen regimes can maintain productivity under DI conditions. Together, these studies

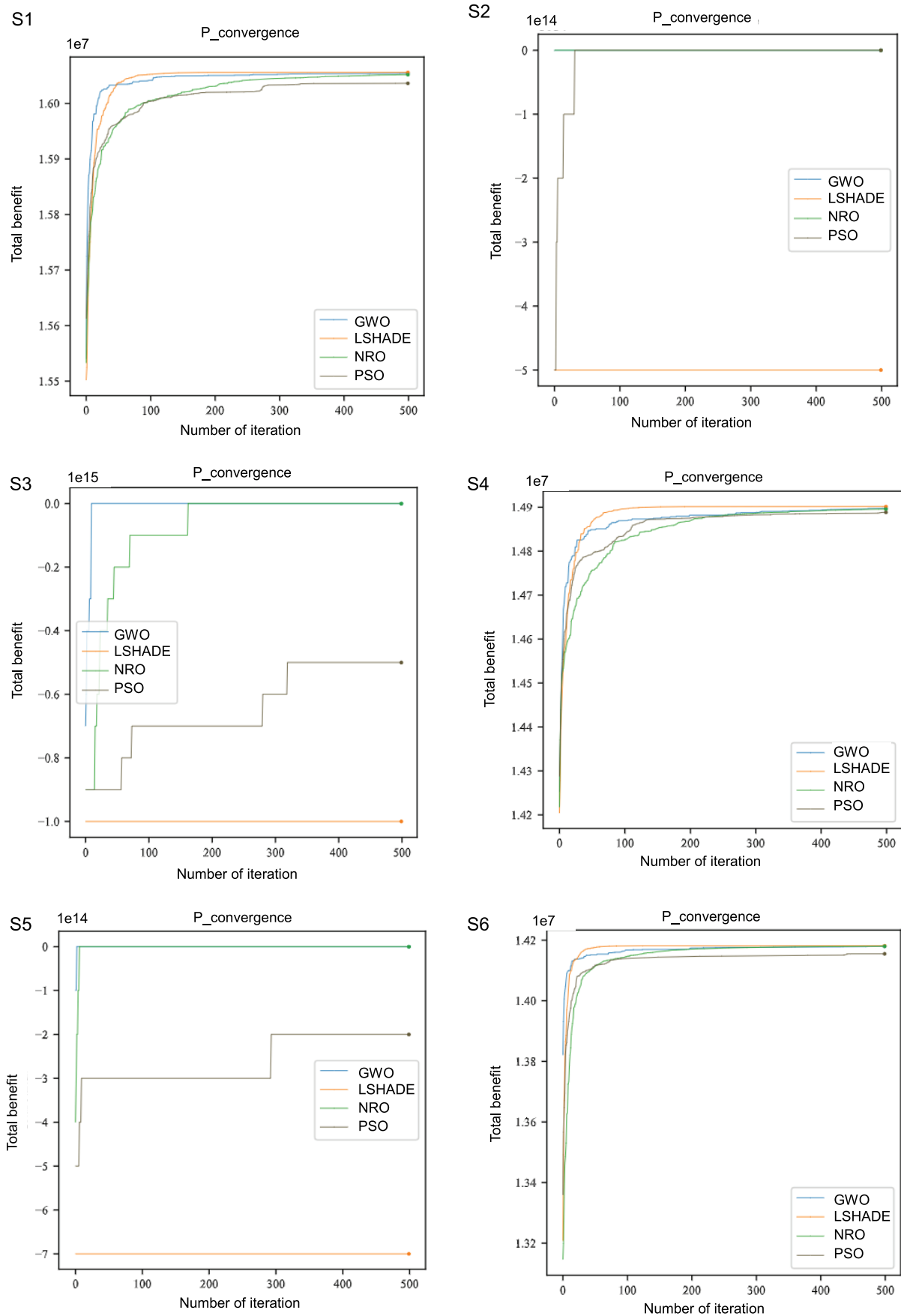


Fig. 2. Convergence plots of the algorithms (GWO: Grey Wolf Optimiser; LSHADE: Success-History based Adaptive Differential Evolution with Linear Population; NRO: Nuclear Reaction Optimisation; PSO: Particle Swarm Optimisation) for six summer scenarios (S1–S6); source: own study

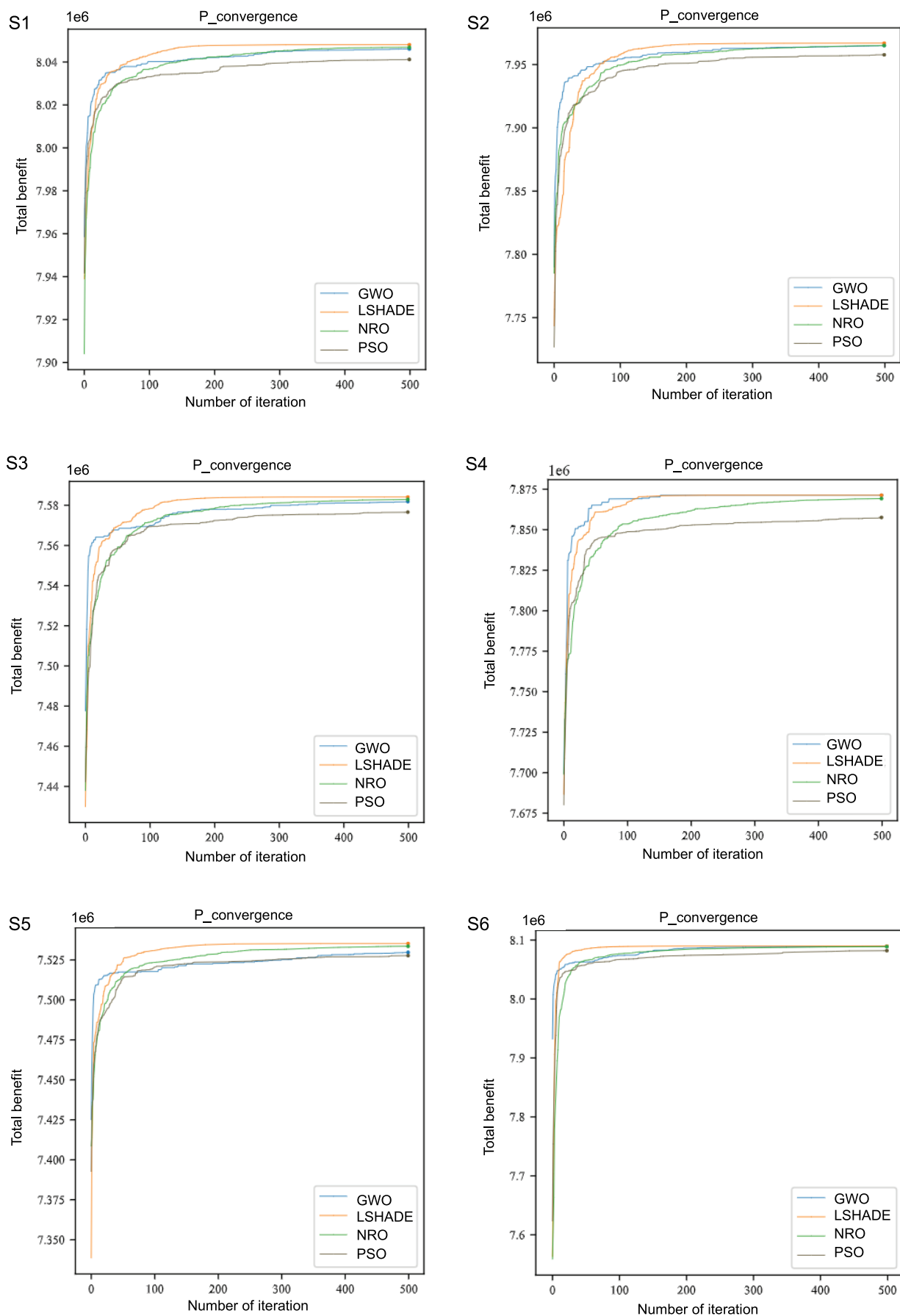


Fig. 3. Convergence plots of the algorithms (GWO: Grey Wolf Optimiser; LSHADE: Success-History based Adaptive Differential Evolution with Linear Population; NRO: Nuclear Reaction Optimisation; PSO: Particle Swarm Optimisation) for six winter scenarios (S1–S6); source: own study

Table 4. Comparison of metaheuristic algorithms for irrigation optimisation in the winter season

Algorithm	Total benefit (USD)			Standard deviation benefit	DI (%)		
	average	maximum	minimum		average	maximum	minimum
LSHADE	6,956.03	6,956.03	6,956.03	0	12.5	12.5	12.5
GWO	6,955.18	6,955.43	6,954.44	347	27.44	32.09	18.64
NRO	6,949.35	6,955.23	6,940.46	6295	28.20	36.98	12.89
PSO	6,947.64	6,954.36	6,939.89	8109	28.05	36.91	12.75

Explanations: GWO = Grey Wolf Optimiser, LSHADE = Success-History based Adaptive Differential Evolution with Linear Population, NRO = Nuclear Reaction Optimisation, PSO = Particle Swarm Optimisation.

Source: own study.

Table 5. Optimal results of crops distribution using optimisation algorithms

Season	Scenario	Algorithm	Total benefit (USD)	Used area (%)	Average DI (%)	Crop pattern (%)
Winter	S1	LSHADE	6,920.10	100.00	12.50	wheat – 45, barley – 24, potato – 5, spinach – 1, alfalfa – 10, forage mixture – 15
	S2	LSHADE	6,850.38	100.00	12.50	wheat – 40, barley – 22, potato – 5, spinach – 3, alfalfa – 10, forage mixture – 20
	S3	LSHADE	6,521.20	100.00	12.50	wheat – 25, barley – 18, potato – 5, spinach – 2, alfalfa – 20, forage mixture – 30
	S4	LSHADE	6,768.13	100.00	12.50	wheat – 35, barley – 25, potato – 10, spinach – 5, alfalfa – 10, forage mixture – 15
	S5	LSHADE	6,479.16	100.00	12.50	wheat – 20, barley – 10, potato – 30, spinach – 20, alfalfa – 10, forage mixture – 10
	S6	LSHADE	6,956.03	100.00	12.50	wheat – 45, barley – 30, potato – 5, spinach – 3, alfalfa – 7, forage mixture – 10
Summer	S1	GWO	13,419.06	83.41	10.67	maize – 20, tomato – 10, cucumber – 15, aubergine – 5, watermelon – 13.41, maize forage – 20
	S2	LSHADE	13,805.45	89.23	12.50	maize – 25, tomato – 5, cucumber – 5, aubergine – 4.23, watermelon – 10, maize forage – 40
	S3	LSHADE	12,812.96	81.08	12.50	maize – 26.08, tomato – 10, cucumber – 10, aubergine – 5, watermelon – 15, maize forage – 15
	S4	GWO	12,119.32	73.48	21.77	maize – 10, tomato – 20, cucumber – 18.46, aubergine – 10, watermelon – 10, maize forage – 5.01
	S5	LSHADE	12,193.98	78.69	12.50	maize – 35, tomato – 8.69, cucumber – 5, aubergine – 5, watermelon – 10, maize forage – 15
	S6	GWO	12,925.03	81.89	8.89	maize – 25, tomato – 10, cucumber – 10, aubergine – 5, watermelon – 11.89, maize forage – 20

Explanations: GWO, LSHADE, NRO, PSO as in Tab. 4.

Source: own study.

reinforce the effectiveness of S6 as a sustainable distribution strategy.

Although a formal uncertainty analysis was not performed, it is important to note that climate projections, crop parameters, and economic data may influence the model results. However, the consistent performance of the optimisation algorithms across different DI scenarios suggests that the proposed framework is reasonably robust. Future research could include sensitivity or stochastic analysis to further examine uncertainty.

CONCLUSIONS

The optimisation algorithms were used in simulations aimed at increasing productivity and improving water use under water-scarce conditions. The optimisation model incorporated multi-objective functions that account for economic returns and resource constraints. Yield reduction (YR) varied because of differences in plant physiological and morphological characteristics, and the sensitivity of individual crops to deficit irrigation

(DI) varied according to growth stage. Sensitive crops, such as potatoes, spinach, and alfalfa in winter and cucumbers, tomatoes, and melons in summer, have shallow roots and high transpiration rates. They also pass through water-demanding stages, such as tuber formation or fruiting, which lead to greater yield losses under DI. In contrast, tolerant crops such as wheat, barley, forage mixture, aubergine, maize, and maize forage have deeper roots, thick or waxy leaves, lower transpiration rates, and sometimes shorter growth cycles, enabling them to maintain productivity under DI conditions. The findings of this paper indicate the importance of balanced crop selection. Scenarios optimised using the Success-History based Adaptive Differential Evolution with Linear Population (LSHADE) algorithm favoured increased allocation to maize and forage crops in the summer season. This improved overall profitability compared with current practices. In winter, prioritising wheat and barley ensured food security and stable income under moderate DI levels. The results demonstrated that integrating AI, particularly the LSHADE algorithm, played a crucial role in identifying the best scenarios for managing DI during the summer and winter seasons. In the summer season, S2, consisting of maize, tomato, cucumber, aubergine, watermelon, and maize forage, produced a 29.09% increase in profit. The superior performance of S2 in the summer season was due to the higher allocation of maize and maize forage, which demonstrated greater tolerance to DI while maintaining stable productivity. This balanced crop distribution reduced yield losses in sensitive vegetables and optimised water-use efficiency, resulting in a significant increase in net profit.

In winter, S6 achieved a 43.96% increase in profits. This scenario relied on essential crops such as wheat, barley, potatoes, spinach, alfalfa, and fodder mixture. The strong performance of S6 was mainly due to the higher allocation of wheat and barley, which showed stable yields under DI. The inclusion of limited high-value crops further enhanced profitability while maintaining efficient land and water use. This diversity demonstrates the potential to achieve both food security and economic sustainability. These results confirm that AI provides an effective means of allocating water resources and agricultural land to maximise profits despite constraints, insufficient irrigation, and climate change. Therefore, this paper recommends adopting this smart approach to agricultural planning to achieve sustainable development goals and ensure food security in this water-scarce region.

CONFLICT OF INTERESTS

All authors declare that they have no conflict of interest.

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